

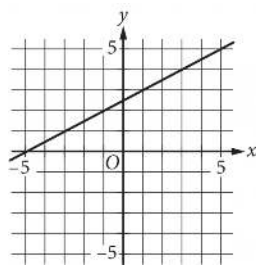
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

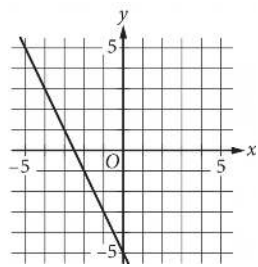
Which of the following is the graph of the equation $y = 2x - 5$ in the xy-plane?

Answer

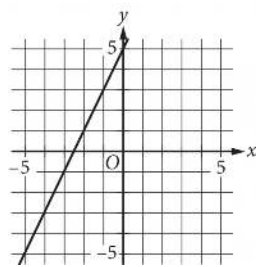
A.



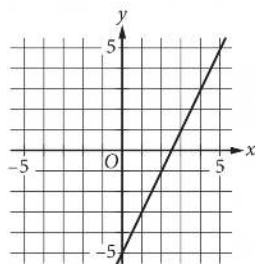
B.



C.



D.



Correct Answer: D

Rationale

Choice D is correct. In the xy -plane, the graph of the equation $y = mx + b$, where m and b are constants, is a line with slope m and y -intercept $(0, b)$. Therefore, the graph of $y = 2x - 5$ in the xy -plane is a line with slope 2 and a y -intercept $(0, -5)$. Having a slope of 2 means that for each increase in x by 1, the value of y increases by 2. Only the graph in choice D has a slope of 2 and crosses the y -axis at $(0, -5)$. Therefore, the graph shown in choice D must be the correct answer.

Choices A, B, and C are incorrect. The graph of $y = 2x - 5$ in the xy -plane is a line with slope 2 and a y -intercept at $(0, -5)$. The graph in choice A crosses the y -axis at the point $(0, 2.5)$, not $(0, -5)$, and it has a slope of $\frac{1}{2}$, not 2. The graph in choice B crosses the y -axis at $(0, -5)$; however, the slope of this line is -2 , not 2. The graph in choice C has a slope of 2; however, the graph crosses the y -axis at $(0, 5)$, not $(0, -5)$.